



COURSE SLIDES · NON-WSQ COURSE

Autodesk Revit Architecture Masterclass

Course Code: C508

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v1.0 · 1 October 2026

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg



COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE

Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE

BIM, CAD and digital design — Revit, AutoCAD, Fusion 360 and engineering software.

DELIVERS

Courses on architecture drawing, BIM modeling, technical drawing and product design.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name and organisation / role.
- Your experience with CAD, BIM or architectural drafting (if any).
- What you want to design or document after this course.



Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

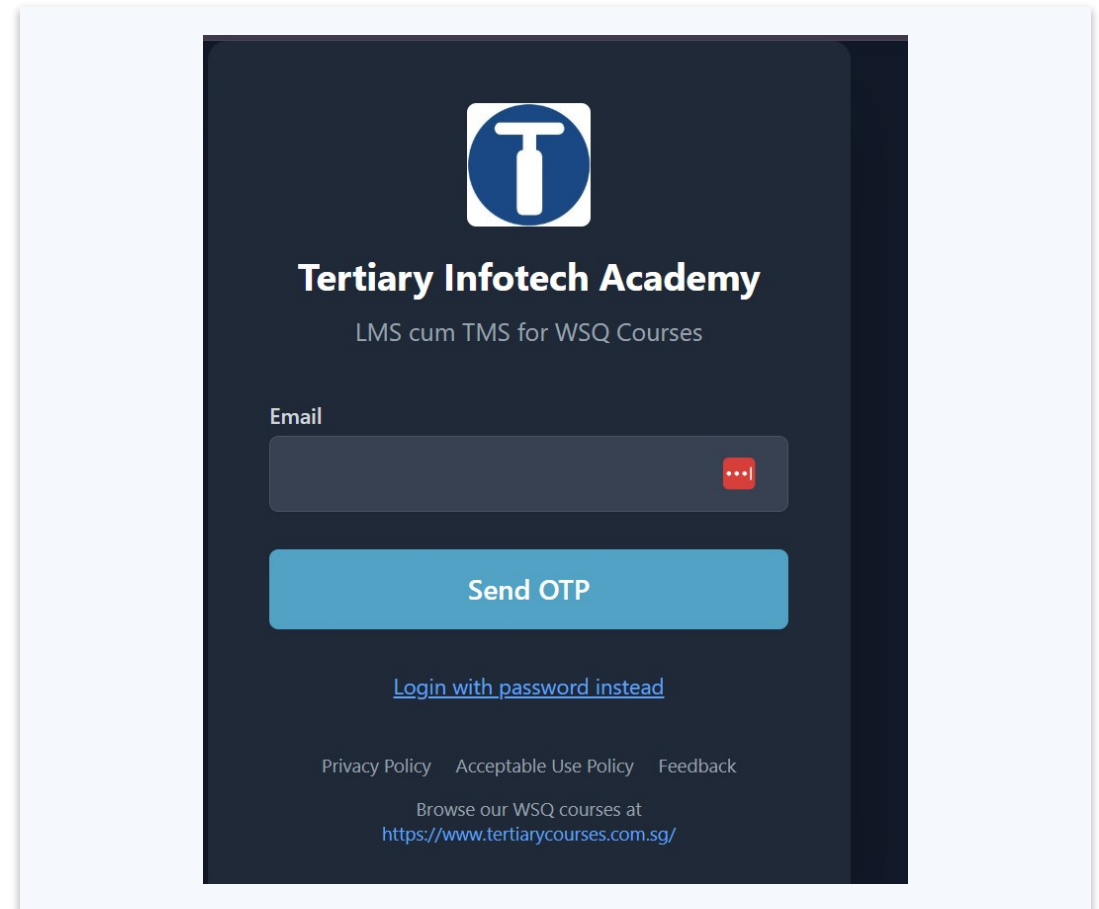
Be punctual; return from breaks on time.

6

Attend both days in full.

Download Course Material

- Log in to the LMS/TMS portal: <https://lms-tms.tertiaryinfotech.com>
- Open your course under My Courses.
- Download the slides, Learner Guide and lab files from the Course Materials section.
- Keep them handy — refer to the slides and Learner Guide during the labs.
- Your certificate of completion is also issued on this portal after the course.



SCHEDULE

Lesson Plan — 2 Days, 7.5 hours/day

Day 1

- Day 1 — Design Modeling & Building Elements
 - Welcome · introductions · course overview
 - Topic 1: Design Modeling in Revit (Labs 1–4)
 - Topic 2: Part Lists for Modeling (Labs 5–8)
 - Day 1 recap & Q&A

Day 2 & timing

- Day 2 — Documentation & Specification
 - Topic 3: Model Documentation (Labs 9–11)
 - Topic 4: Design Specification (Lab 12)
 - Course summary & feedback
- Daily timing
 - 9:30am–5:30pm · 30-min lunch



REVIT & BIM FUNDAMENTALS

One Model, Every Drawing

What is Revit? What is BIM?

1

Design & documentation platform

Revit supports the design, drawings and schedules required for building information modeling (BIM).

2

Information when you need it

BIM delivers information about project design, scope, quantities and phases on demand.

3

One virtual building model

Every drawing sheet, 2D and 3D view and schedule is a presentation of the same model.

4

Coordinated everywhere

Revit collects information about the project and coordinates it across all other representations.

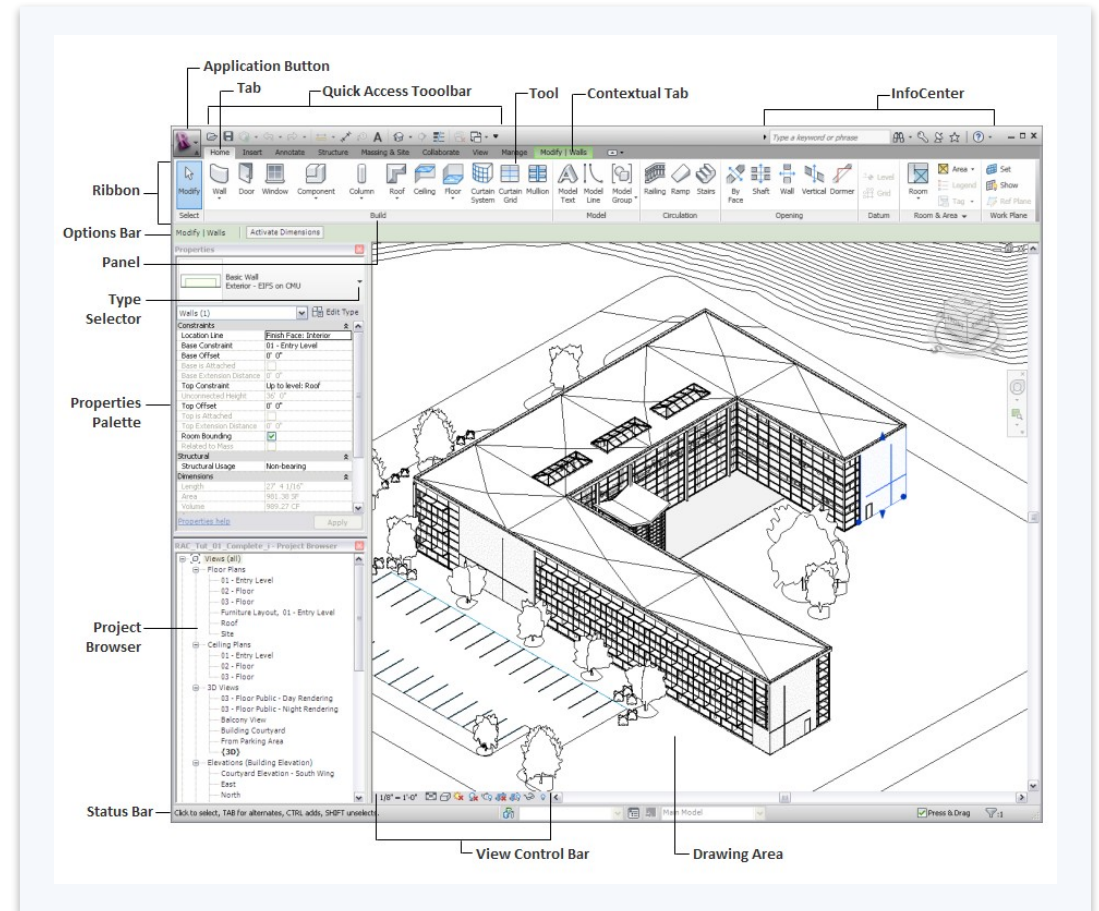
PARAMETRIC CHANGE ENGINE

Change it once — it changes everywhere.

The Revit parametric change engine automatically coordinates changes made anywhere: model views, drawing sheets, schedules, sections and plans.

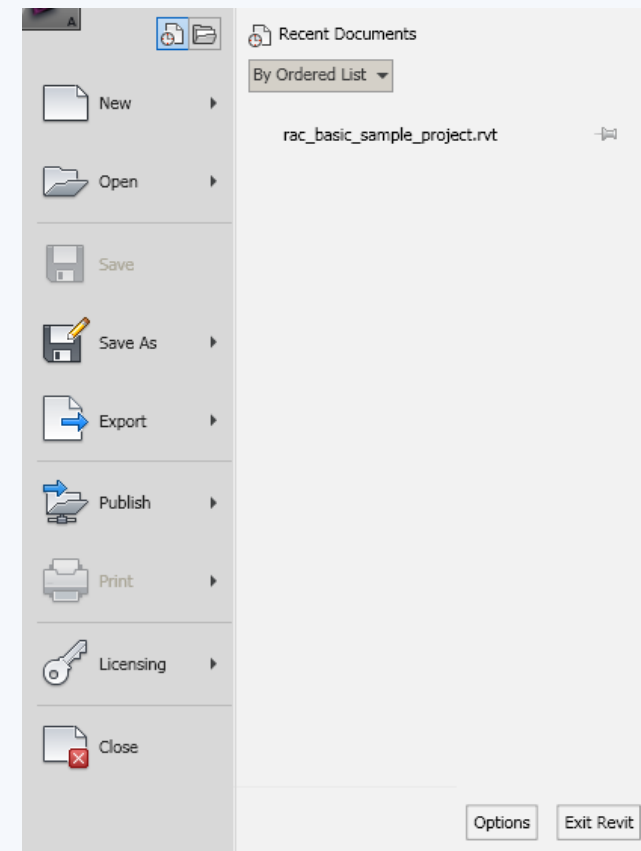
The Revit User Interface

- The interface is designed to simplify your workflow — a few clicks change it to suit how you work.
- The ribbon has 4 display settings; project views can be tiled or layered.
- Core parts: ribbon, Application Menu, Quick Access Toolbar, Project Browser, Properties palette, Options Bar, View Control Bar and status bar.



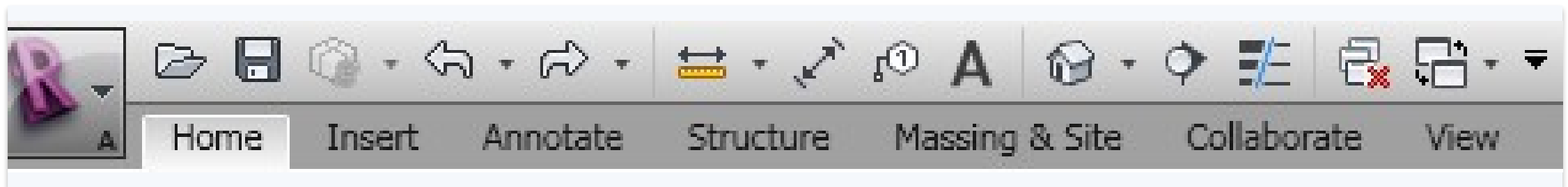
Application Menu

- Access common file actions: New, Open, Save.
- Manage files with advanced tools such as Export and Publish.
- Click the File tab (application button) at the top-left to open it.



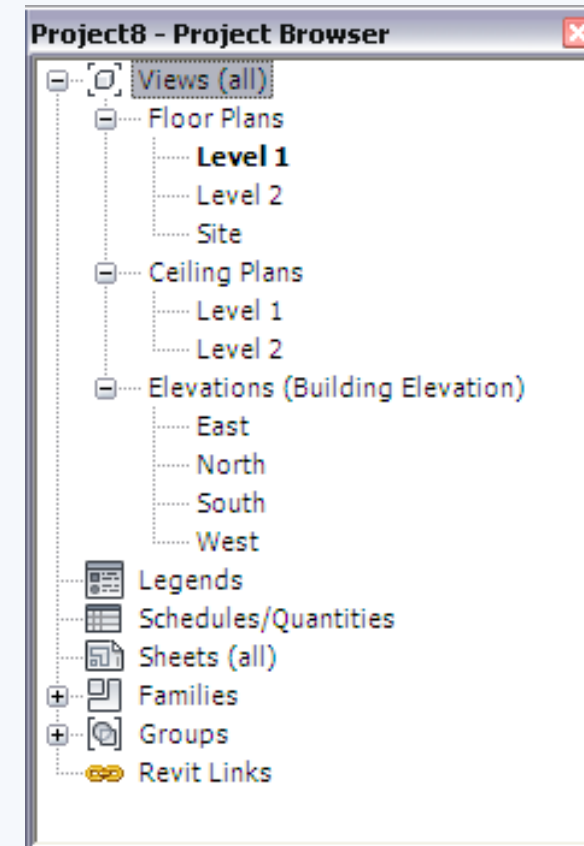
Quick Access Toolbar

- A customisable strip of the tools you use most often.
- Right-click any ribbon tool ▶ Add to Quick Access Toolbar.
- Drop-down at the end: show/hide tools, or move the toolbar below the ribbon.



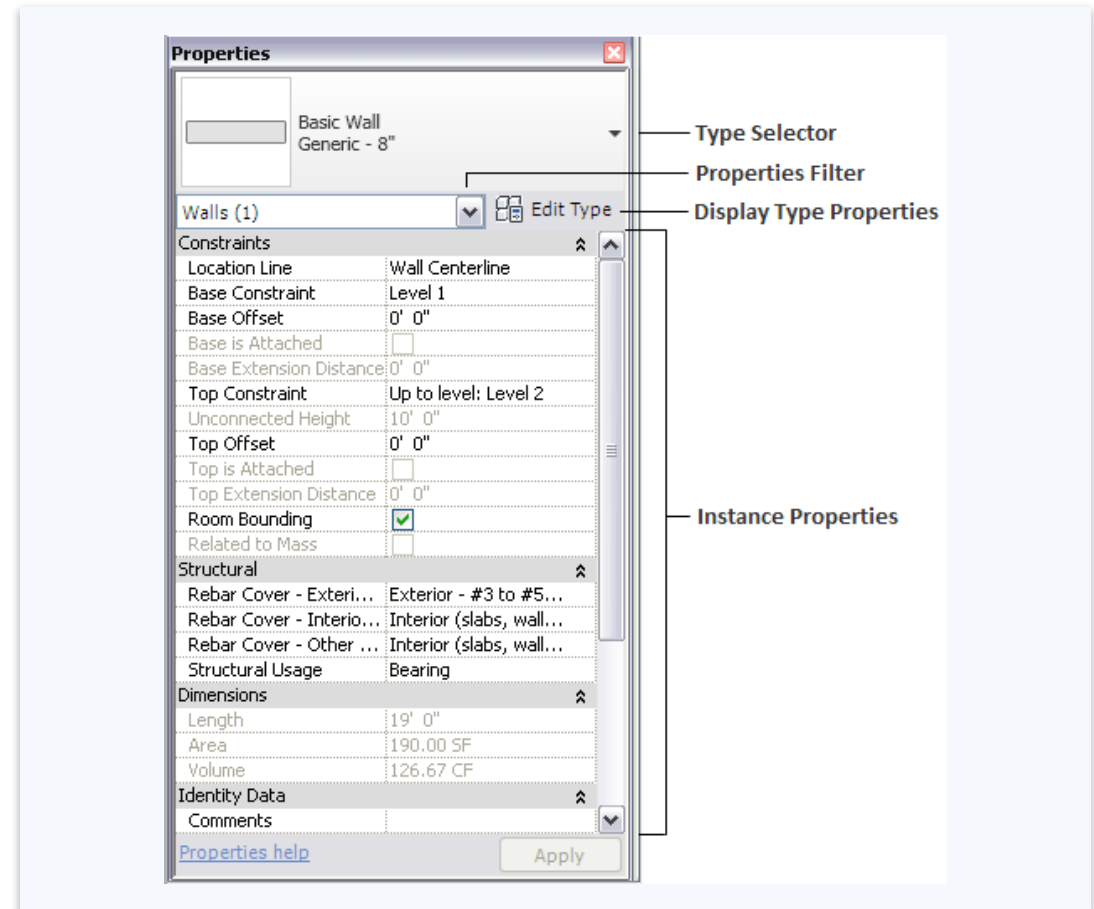
Project Browser

- A logical hierarchy of all views, schedules, sheets, families, groups and linked models in the project.
- Expand/collapse each branch to reach lower-level items.
- Show it via View tab ► Windows panel ► User Interface ► Project Browser.



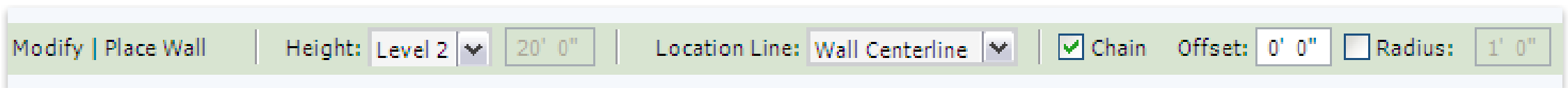
Properties Palette

- A modeless dialog to view and modify the parameters that define each element's properties.
- Top: the Type Selector. Below: instance properties of the selection (or the view).
- Open it via View tab ► Windows panel ► User Interface ► Properties.



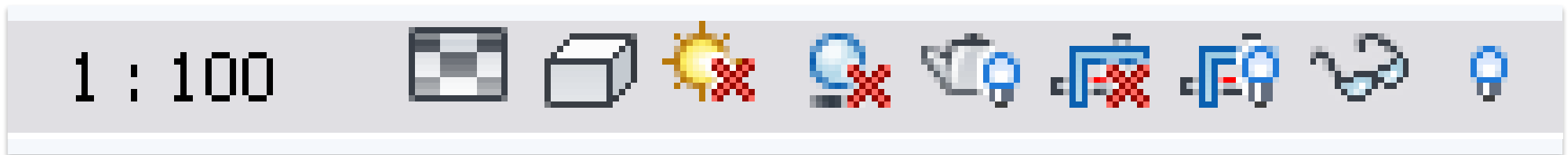
Options Bar

- Sits below the ribbon; its contents change with the current tool or selected element.
- Sets tool options such as Level, Height, Location Line, Chain and Offset while you draw.
- Right-click it to dock at the bottom of the window, above the status bar.



View Control Bar

- Quick access to functions that affect the current view.
- Scale · Detail Level · Visual Style · Sun Path · Shadows · Rendering dialog.
- Crop View · Show/Hide Crop Region · Temporary Hide/Isolate · Reveal Hidden Elements.



Revit File Types

1

RVT — project

The building model: geometry, views, sheets and schedules all live here.

2

RTE — project template

A starting point with preset units, levels, views and families.

3

RFA — loadable family

Doors, windows, furniture, tags and title blocks loaded into projects.

4

DWG / DXF — CAD data

AutoCAD drawings that can be linked or imported into the model.

TOPIC 01

Design Modeling in Revit

BIM & the Revit platform · User interface · Site design & toposurfaces · Massing · Materials · Rendering

Key Concepts — Design Modeling in Revit

1

BIM platform

Revit supports design, drawings and schedules for building information modeling — every sheet, view and schedule presents the same virtual building model.

2

Parametric change engine

A change made anywhere — model views, sheets, schedules, sections, plans — is coordinated automatically everywhere else.

3

Revit user interface

Ribbon, Application Menu, Quick Access Toolbar, Project Browser, Properties palette, Options Bar and View Control Bar.

4

Site design

Sketch a toposurface, then add property lines, a building pad, parking and site components; present it in 3D or rendered views.

5

Massing studies

Conceptual mass families and in-place masses let you study volumes, zoning and floor-area ratio before detailed modeling.

6

Materials & rendering

Materials control graphics, appearance, thermal and physical data; the Render tool produces photorealistic images of the model.

Hands-On Labs — Design Modeling in Revit

Labs 1–2

- Create a Project and Add Levels
- Create a Terrain and Building Pad

Labs 3–4

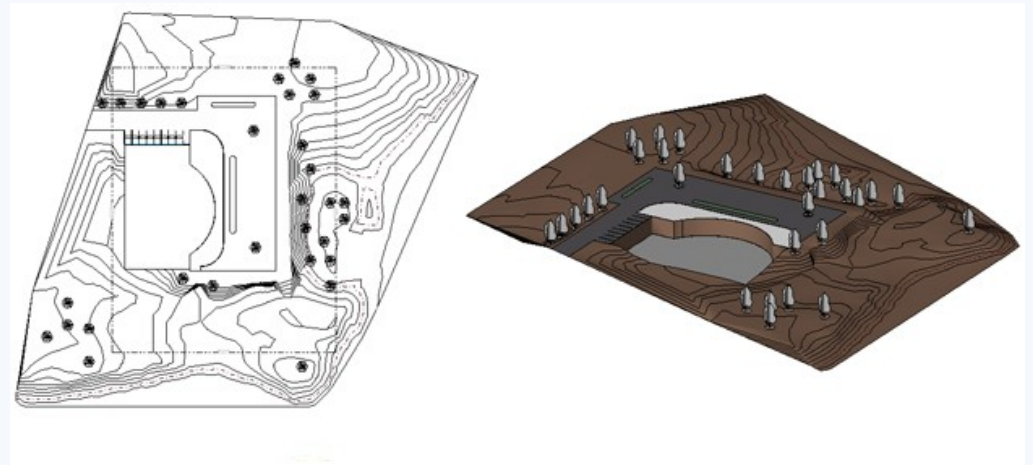
- Create an In-Place Mass and Apply Materials
- Render a Photorealistic View

—

- —

Site Design

- Revit provides tools to lay out a complete site plan.
- Start by sketching a toposurface, then add property lines, a building pad, parking and site components.
- Present the site as a 3D view or render it for a realistic presentation.



Toposurface Toolset (Massing & Site tab)

1

Toposurface

Defines a topographical surface from points or imported data, in 3D or site-plan views.

2

Subregion

An area of the toposurface with its own properties, such as a different material.

3

Split / Merge Surface

Split a surface into two independent surfaces, or rejoin overlapping surfaces.

4

Graded Region

Grade the terrain: Revit demolishes the original surface and creates the proposed copy.

5

Building Pad

A closed-loop pad cut into the toposurface, with height offset, slope and openings.

6

Point elevation modes

Absolute Elevation places points at a set height; Relative to Surface edits an existing surface.

Massing — Two Ways to Study a Volume

Family-based

- Conceptual Mass Family
 - Created outside the project and loaded in
 - Reusable across projects and design options
 - Best for often-used building volumes

In-place

- In-Place Mass
 - Created inside the current project
 - Specific to this project's context — not reusable
 - Generate floors, roofs, curtain systems and walls from the mass

What Materials Define

1

Graphics

Shading colour, surface pattern and cut pattern in non-rendered views.

2

Appearance

The photorealistic look used by the Realistic style and the renderer.

3

Thermal & Physical

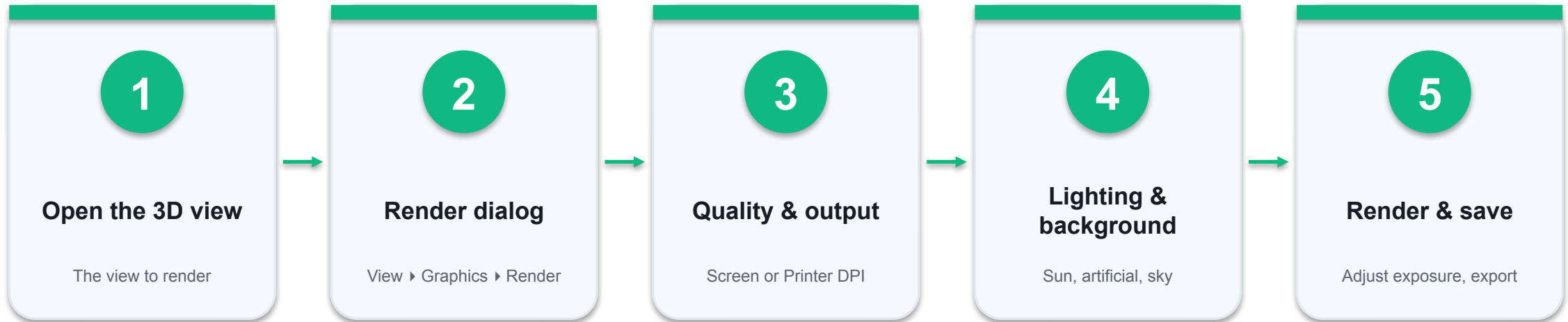
Analytical data for energy and structural analysis.

4

How to apply

By category, by family, by element parameter, or by face with the Paint tool.

Rendering Workflow



TIP

Use the Realistic visual style for a live preview; render only when the view, materials and lighting are ready — high quality renders take time.

Create a Project and Add Levels

LAB 1

Maps to: produce and visualise 2D/3D technical drawings using Revit (LO1)

Start a new Revit project from the architectural template, rename the default levels and add new levels that become the vertical datums for the whole building model.

YOU'LL BUILD

A Revit project with four named, correctly-elevated levels and their floor-plan views.

TOOLCHAIN

Revit · Architectural Template · Level tool

DONE WHEN

The Project Browser lists a floor plan for each new level, and the South elevation shows all four levels with the correct names and elevations. Compare with the completed LittleHouse.rvt model in the lab folder.

Create a Project and Add Levels — Steps 1–3

1

Start Revit and create a new project from the architectural template.

File ▶ New ▶ Project ▶ Architectural Template ▶ OK

2

Open a building elevation so the levels are visible.

Project Browser ▶ Elevations (Building Elevation) ▶ South

3

Rename Level 1 to '01 - Entry Level'; accept renaming the corresponding views.

Double-click the level name ▶ type 01 - Entry Level ▶ Enter ▶ Yes

Create a Project and Add Levels — Steps 4–6

4

Rename Level 2 to '02 - Upper Level' and set its elevation by clicking the blue elevation value.

Double-click the elevation value ▶ type 3000 ▶ Enter

5

Add a roof level by drawing a new level above the existing ones.

Architecture ▶ Datum ▶ Level ▶ draw left to right above 02

6

Add another level by offsetting from an existing one, then rename it.

Modify | Place Level ▶ Pick Lines ▶ Offset 3000 ▶ click level 02

Create a Project and Add Levels

✓ Expected result

The Project Browser lists a floor plan for each new level, and the South elevation shows all four levels with the correct names and elevations. Compare with the completed LittleHouse.rvt model in the lab folder.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Create a Terrain and Building Pad

LAB 2

Maps to: produce and visualise 2D/3D technical drawings using Revit (LO1)

Add a toposurface to the building site by placing points at different elevations, then cut a building pad into the terrain based on the footprint of the foundation walls.

YOU'LL BUILD

A landscaped site with a graded toposurface, a grass material and a building pad ready for the model.

TOOLCHAIN

Revit · Toposurface · Building Pad · Section Box

DONE WHEN

The 3D view shows the terrain with a grass material, and the building pad cuts a level recess into the toposurface at the foundation walls.

Create a Terrain and Building Pad — Steps 1–3

1

Open the site plan view of the project.

Project Browser ▶ Floor Plans ▶ Site

2

Start the Toposurface tool and place boundary points at elevation 0.

Massing & Site ▶ Model Site ▶ Toposurface ▶ Place Point ▶ Elevation 0

3

Raise the far points: change the elevation on the Options Bar as you place them.

Options Bar ▶ Elevation 900 ▶ click to place the back points

Create a Terrain and Building Pad — Steps 4–6

4

Assign a grass material to the surface, then finish it.

Properties ▶ Material ▶ Grass ▶ Finish Surface

5

Sketch a building pad over the foundation footprint on the lower level.

Massing & Site ▶ Model Site ▶ Building Pad ▶ Pick Walls ▶ Finish

6

Frame the result in 3D with a section box to see the pad cutting the terrain.

3D view ▶ Properties ▶ Section Box ✓ ▶ drag the handles

Create a Terrain and Building Pad

✓ Expected result

The 3D view shows the terrain with a grass material, and the building pad cuts a level recess into the toposurface at the foundation walls.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Create an In-Place Mass and Apply Materials

LAB 3

Maps to: produce and visualise 2D/3D technical drawings using Revit (LO1)

Model a conceptual building volume as an in-place mass, then use the Materials browser to control how model elements display in shaded and rendered views.

YOU'LL BUILD

An in-place conceptual mass with materials applied by category and by face.

TOOLCHAIN

Revit · In-Place Mass · Materials Browser · Paint tool

DONE WHEN

The mass displays the painted material on the selected face in a shaded 3D view, and the duplicated material appears in the project's material library.

Create an In-Place Mass and Apply Materials — Steps 1–3

1

Start an in-place mass and name the new mass family.

Massing & Site ▶ Conceptual Mass ▶ In-Place Mass ▶ name it Tower ▶ OK

2

Draw a closed profile and create a form from it.

Draw ▶ Rectangle ▶ select the profile ▶ Modify ▶ Create Form ▶ Solid Form

3

Finish the mass to return to the project environment.

In-Place Editor ▶ Finish Mass

Create an In-Place Mass and Apply Materials — Steps 4–6

4

Open the Materials browser and duplicate a material to customise it.

Manage ▶ Settings ▶ Materials ▶ right-click Concrete ▶ Duplicate

5

Set the material's graphics: shading colour, surface pattern and cut pattern.

Material Editor ▶ Graphics ▶ Shading ▶ choose a colour ▶ OK

6

Paint the new material onto one face of the mass.

Modify ▶ Geometry ▶ Paint ▶ pick the material ▶ click a face

Create an In-Place Mass and Apply Materials

✓ Expected result

The mass displays the painted material on the selected face in a shaded 3D view, and the duplicated material appears in the project's material library.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Render a Photorealistic View

LAB 4

Maps to: produce and visualise 2D/3D technical drawings using Revit (LO1)

Present the design by producing a photorealistic image of the building model: set the render quality, resolution, lighting and background, then export the image.

YOU'LL BUILD

A rendered exterior image of the LittleHouse model, exported to the project and saved as a file.

TOOLCHAIN

Revit · Render dialog · Realistic visual style

DONE WHEN

A photorealistic rendering of the house appears under Project Browser ▶ Renderings, and the exported PNG opens outside Revit.

Render a Photorealistic View — Steps 1–3

1

Open LittleHouse.rvt from the lab folder and open the default 3D view.

Quick Access Toolbar ▶ Default 3D View

2

Preview materials in real time with the Realistic visual style.

View Control Bar ▶ Visual Style ▶ Realistic

3

Open the Rendering dialog from the View tab.

View ▶ Graphics ▶ Render

Render a Photorealistic View — Steps 4–6

4

Set Quality to Medium and Output to Screen resolution.

Rendering dialog ▶ Quality: Medium ▶ Resolution: Screen

5

Set the lighting scheme and a sky background, then render.

Lighting: Exterior Sun only ▶ Background: Sky ▶ Render

6

Adjust exposure if needed, then save the image into the project and export it.

Adjust Exposure ▶ Save to Project ▶ Export ▶ littlehouse-render.png

Render a Photorealistic View

✓ Expected result

A photorealistic rendering of the house appears under Project Browser ► Renderings, and the exported PNG opens outside Revit.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Recap — Design Modeling in Revit

- You can now: produce and visualise 2D/3D technical drawings using Revit (Lab 1: Create a Project and Add Levels)
- You can now: produce and visualise 2D/3D technical drawings using Revit (Lab 2: Create a Terrain and Building Pad)
- You can now: produce and visualise 2D/3D technical drawings using Revit (Lab 3: Create an In-Place Mass and Apply Materials)
- You can now: produce and visualise 2D/3D technical drawings using Revit (Lab 4: Render a Photorealistic View)

TOPIC 02

Part Lists for Modeling in Revit

Levels & grids · Stacked, basic and curtain walls · Doors & windows · Roofs · Ceilings · Floors & openings · Stairs · Columns

Key Concepts — Part Lists for Modeling in Revit

1

Levels & grids

Levels are datums for each storey or reference height; grids are annotation elements that organise the design — add columns at grid intersections.

2

Wall types

Basic walls, stacked walls (two or more subwalls of different thickness) and curtain walls with grids, panels and mullions.

3

Doors & windows

Hosted components that automatically cut their opening; press Spacebar to flip the swing before placement.

4

Roofs & ceilings

Create roofs by footprint, extrusion, sloped glazing or from a mass; place ceilings automatically inside walls or by sketch.

5

Floors & openings

Sketch floor boundaries as closed loops; cut openings By Face, Vertical, or through the full height with the Shaft tool.

6

Stairs & railings

Assemble runs, landings, supports and railings in stair assembly edit mode, in plan or 3D views.

Hands-On Labs — Part Lists for Modeling in Revit

Labs 5–6

- Create Walls
- Place Doors and Windows

Labs 7–8

- Create a Floor and a Roof by Footprint
- Create Stairs and Railings

—

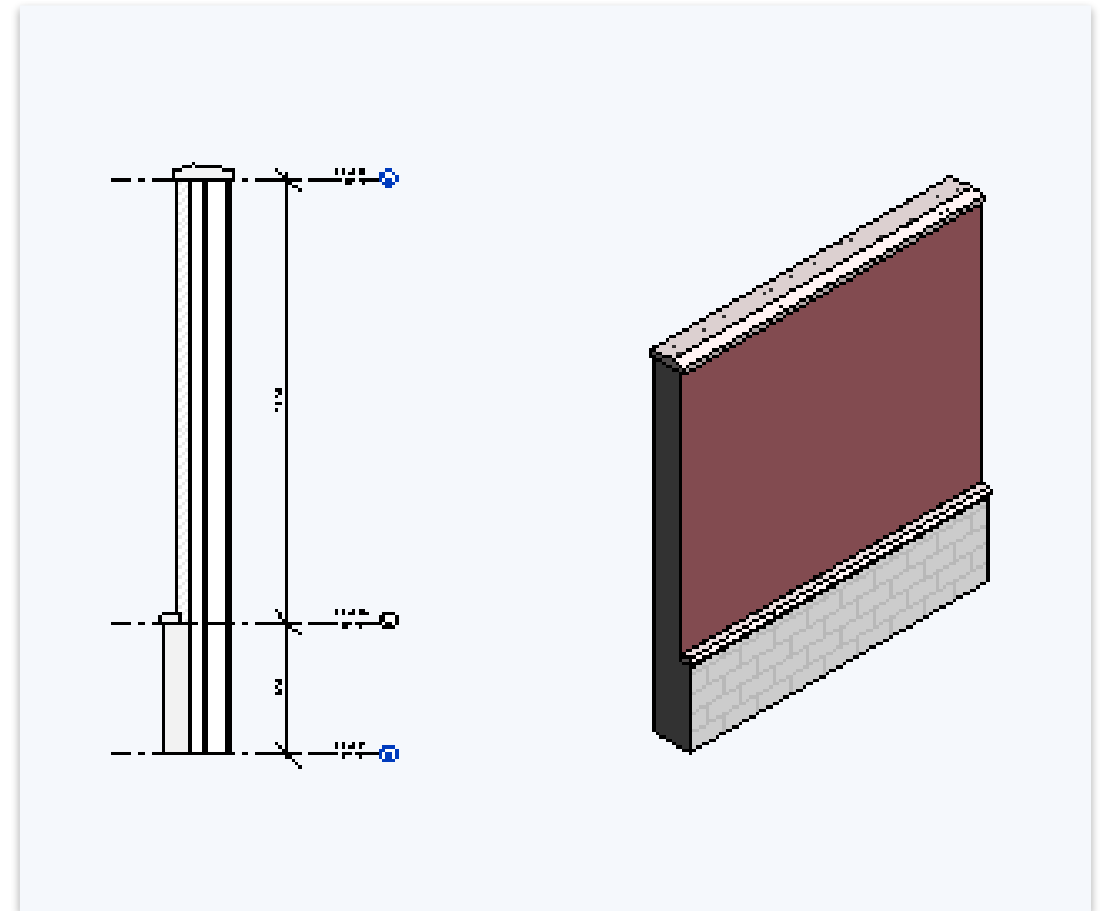
- —

Levels and Grids

- Levels are the vertical datums — create one for each storey plus reference levels such as a sill level.
- Add levels in a section or elevation view: Architecture ▶ Datum ▶ Level.
- Grids organise the design — Architecture ▶ Datum ▶ Grid, with Pick Lines to snap a grid to an existing wall.
- Columns can be placed automatically at every selected grid intersection.

Stacked Walls

- A stacked wall is composed of 2 or more subwalls stacked on top of each other.
- Subwalls can have different thicknesses at different heights.
- All subwalls are attached and their geometry is joined.



Basic Wall Placement Options

1

Level

The base constraint — in 3D views you can choose any level, including non-storey levels.

2

Height

The top constraint level, or an Unconnected height value.

3

Location Line

Which vertical plane follows your cursor: centerline, core, or a face.

4

Chain

Draw a connected series of wall segments end to end.

5

Offset

Place the wall's location line at a distance from the cursor or picked line.

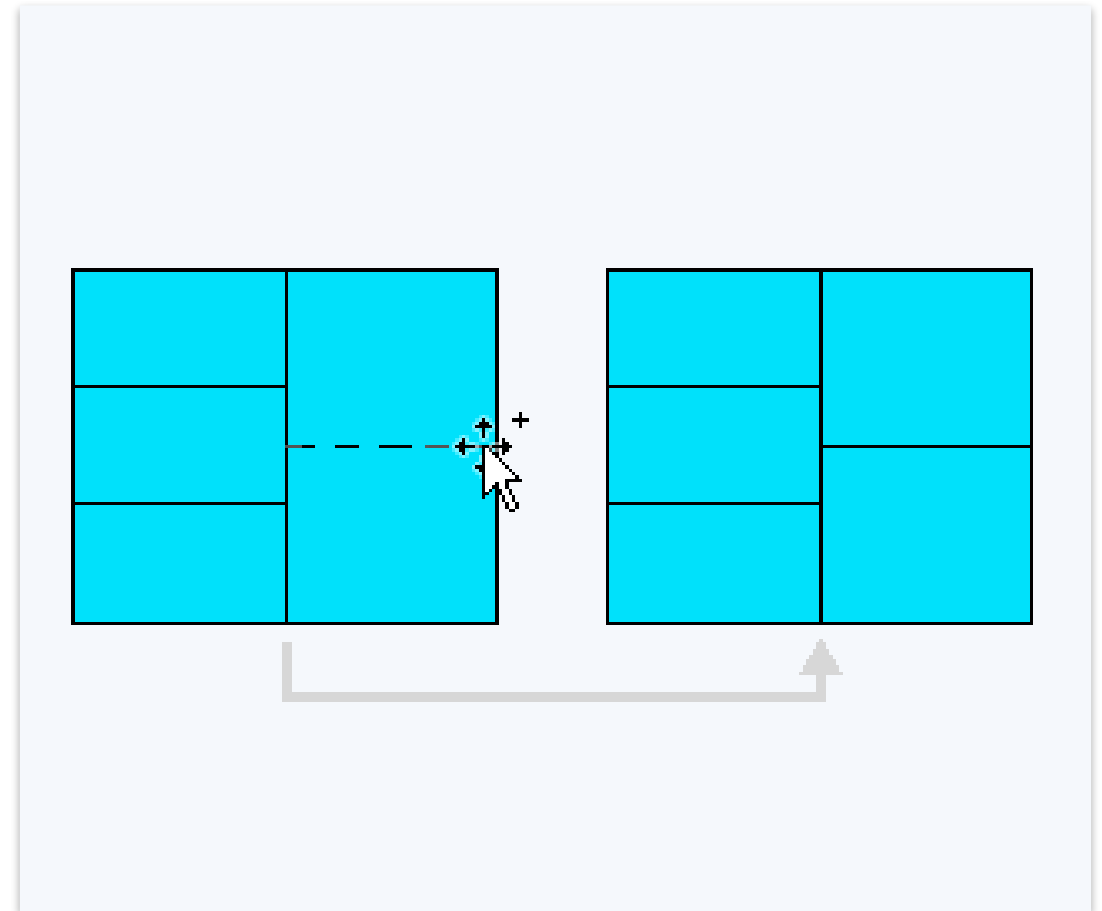
6

Join Status

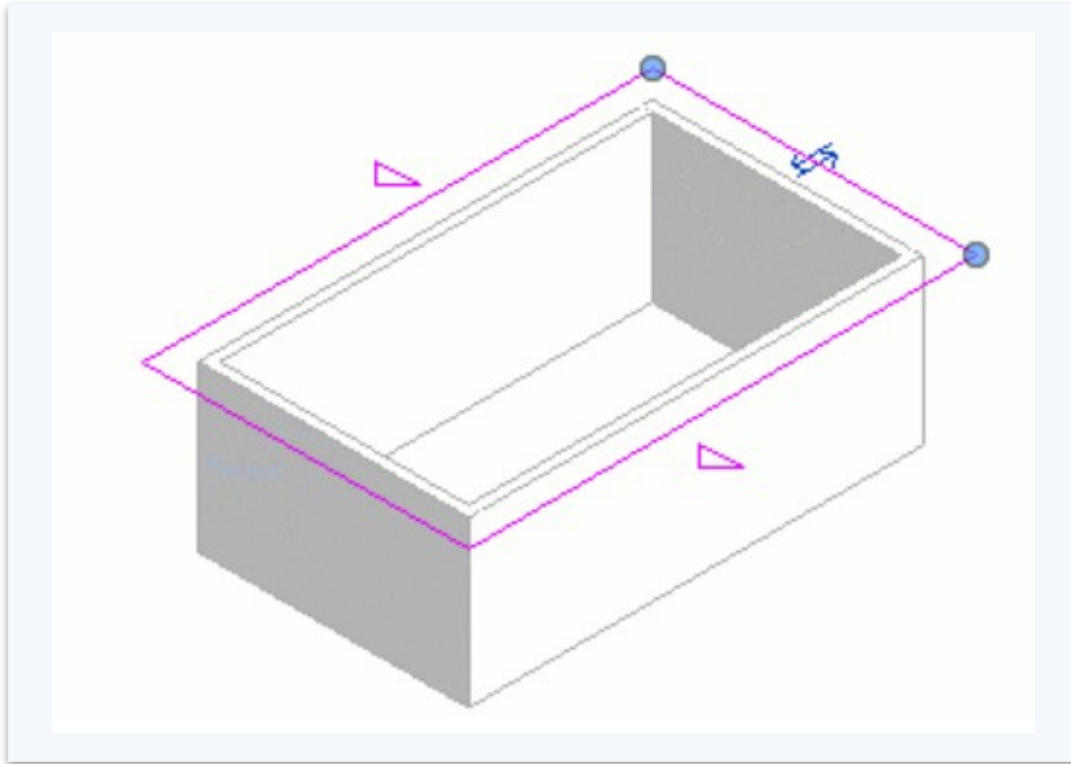
Allow butt joins where walls meet, or Disallow to keep them separate.

Curtain Walls

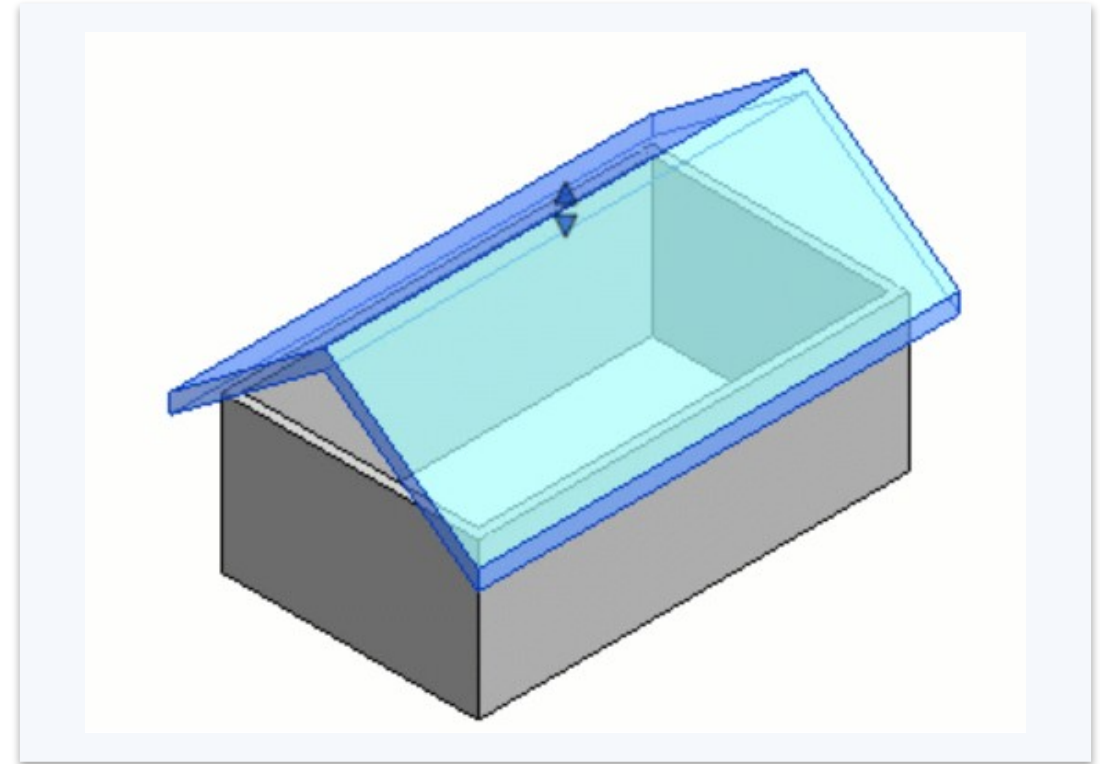
- A curtain wall is placed like any wall — just pick a curtain wall type in the Type Selector.
- After placement, divide the wall with curtain grids, then place mullions on the grid lines.
- Select a panel (Tab to cycle), then swap its type — including doors and walls embedded in panels.
- Architecture ► Build ► Curtain Grid · Add/Remove Segments to customise the layout.



Roof by Footprint

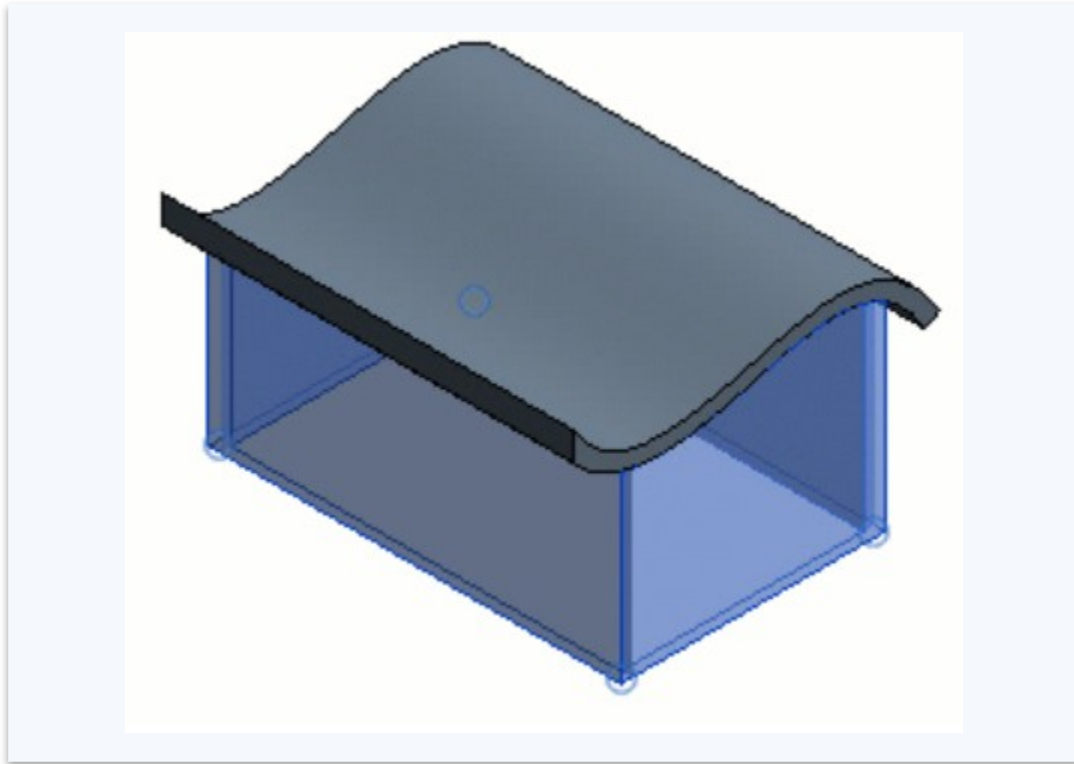


Sketch: a closed loop with slope-defining lines (△ marker)

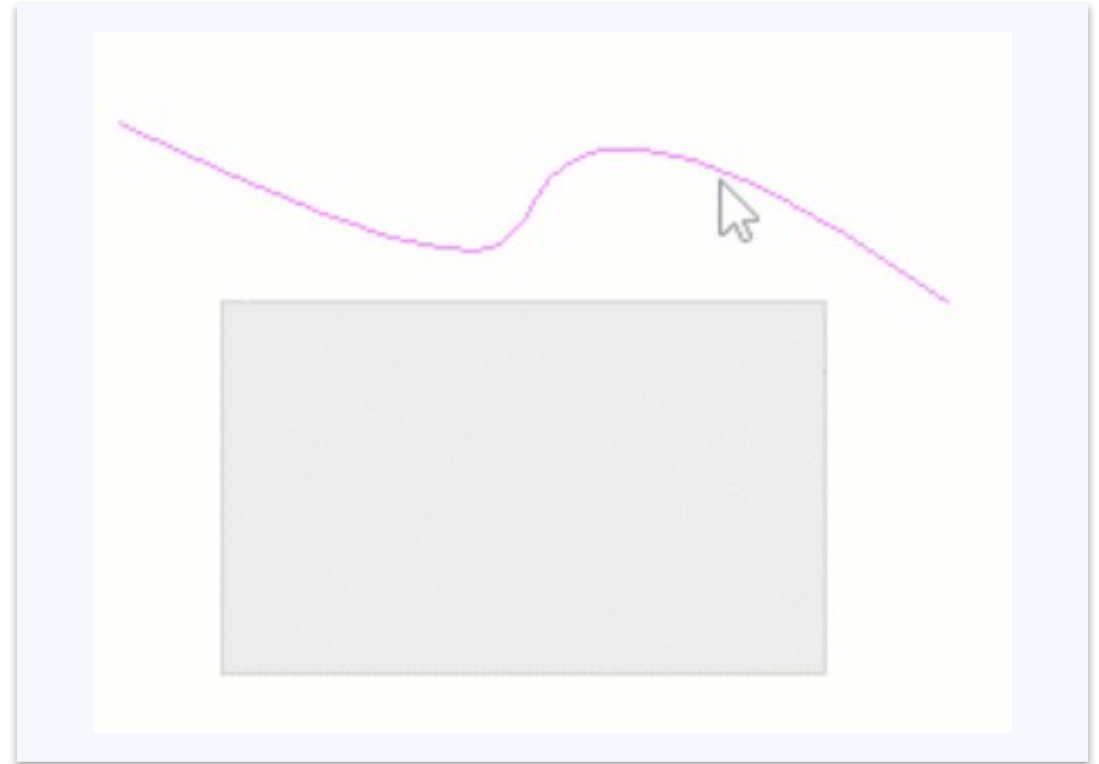


Result: the finished footprint roof in 3D

Roof by Extrusion



Sketch the profile as an open loop in an elevation view



The profile extrudes into the roof; attach walls afterwards

Ceilings, Floors and Openings

1

Ceiling

In a ceiling plan: Automatic Ceiling fills walls that form a closed loop, or sketch the boundary.

2

Floor

Pick walls or sketch a closed-loop profile; extra closed loops inside become openings.

3

Opening — By Face / Vertical

Cut perpendicular to a roof/floor/ceiling face, or vertically to a level.

4

Shaft opening

Cuts through the entire height — every floor, roof and ceiling it crosses, on all levels.

5

Dormer opening

Join the dormer roof to the main roof, then pick the boundary edges to cut the opening.

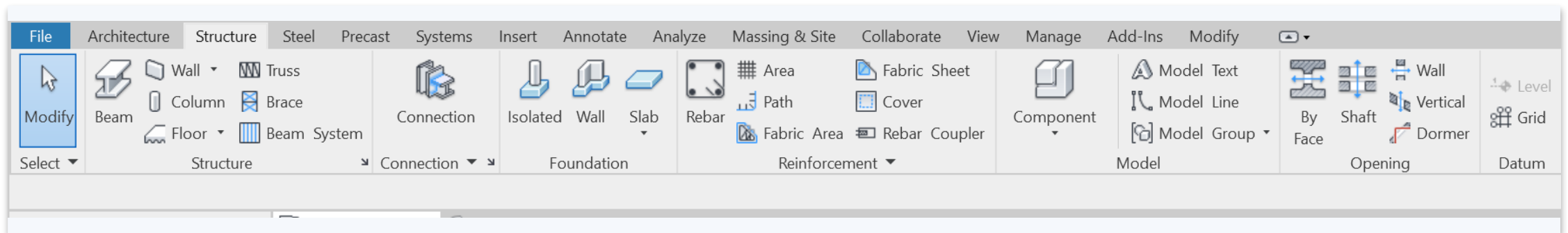
6

Stairs

Assemble runs, landings, supports and railings in stair assembly edit mode — plan or 3D.

Add Columns to Grids

- Structure ► Structure ► Structural Column ► Multiple ► At Grids.
- Select grid lines — a column previews at every intersection.
- Spacebar rotates all the columns being placed; Ctrl-drag to add more intersections.
- Click Finish to create all the columns in one action.



Create Walls

LAB 5

Maps to: create part lists for modeling using Revit (LO2)

Work on different levels to add the exterior walls, interior partition walls and a corridor to the project, then tidy the joins with the Trim/Extend tool.

YOU'LL BUILD

A building shell with exterior walls, interior walls and a corridor opening on the lower level.

TOOLCHAIN

Revit · Wall: Architectural · Trim/Extend

DONE WHEN

The lower level shows a closed exterior shell with interior partitions and a corridor; walls meet cleanly with no gaps or crossing lines. Compare with ModelCreation.rvt in the lab folder.

Create Walls — Steps 1–3

1

Open the lower-level floor plan and start the architectural Wall tool.

Architecture ▶ Build ▶ Wall ▶ Wall: Architectural

2

Pick the wall type and set base/top constraints in the Properties palette.

Type Selector ▶ Basic Wall: Generic 200mm ▶ Top Constraint: Up to level 02

3

On the Options Bar set the Location Line and enable Chain, then draw the exterior walls clockwise.

Options Bar ▶ Location Line: Wall Centerline ▶ Chain ✓

Create Walls — Steps 4–6

4

Add the interior partition walls with a thinner wall type.

Type Selector ▶ Basic Wall: Interior 135mm Partition

5

Create the corridor by trimming the partition walls to form the opening.

Modify ▶ Trim/Extend to Corner ▶ click the two walls to keep

6

Check the shell in 3D and correct any wall join problems.

Default 3D View ▶ inspect ▶ Modify ▶ Wall Joins if needed

Create Walls

✓ Expected result

The lower level shows a closed exterior shell with interior partitions and a corridor; walls meet cleanly with no gaps or crossing lines. Compare with ModelCreation.rvt in the lab folder.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Place Doors and Windows

LAB 6

Maps to: create part lists for modeling using Revit (LO2)

Load door families into the project, then place interior and exterior doors and windows — flipping swing and orientation with the Spacebar and flip controls.

YOU'LL BUILD

A model with doors and windows placed and correctly oriented in the walls, each opening cut automatically.

TOOLCHAIN

Revit · Door tool · Window tool · Load Family

DONE WHEN

Every door and window sits in a wall with its opening cut automatically, tags show sequential numbers, and all swings open the intended way.

Place Doors and Windows — Steps 1–3

1

Open the entry-level floor plan and start the Door tool.

Architecture ▶ Build ▶ Door

2

Load additional door families from the library.

Modify | Place Door ▶ Mode ▶ Load Family ▶ Doors ▶ choose an entrance door

3

Enable tagging on placement so each door is numbered automatically.

Modify | Place Door ▶ Tag ▶ Tag on Placement

Place Doors and Windows — Steps 4–6

4

Hover over a wall, press Spacebar to flip the swing, then click to place the door.

Spacebar to flip ▶ click to place

5

Place windows on the exterior walls with the Window tool.

Architecture ▶ Build ▶ Window ▶ click the exterior walls

6

Fix any wrongly-placed component with the blue flip-control arrows.

Select the door/window ▶ click the flip arrows

Place Doors and Windows

✓ Expected result

Every door and window sits in a wall with its opening cut automatically, tags show sequential numbers, and all swings open the intended way.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Create a Floor and a Roof by Footprint

LAB 7

Maps to: create part lists for modeling using Revit (LO2)

Create a mezzanine floor by sketching its boundary, edit the sketch with Align and Trim into a closed loop, then complete the envelope with a footprint roof whose slopes you define.

YOU'LL BUILD

A mezzanine floor in the store-room area and a sloped roof closing the building envelope.

TOOLCHAIN

Revit · Floor: Architectural · Roof by Footprint · Shaft Opening

DONE WHEN

The mezzanine floor appears with a clean shaft opening, and the 3D view shows a sloped roof sitting on the footprint — no warnings about open loops.

Create a Floor and a Roof by Footprint — Steps 1–3

1

Open the upper-level plan and start the architectural Floor tool.

Architecture ▶ Build ▶ Floor ▶ Floor: Architectural

2

Pick the bounding walls, then switch to Line to close the open edge.

Modify | Create Floor Boundary ▶ Pick Walls ▶ Draw ▶ Line

3

Use Align and Trim to make the sketch one closed loop, then finish.

Modify ▶ Align / Trim ▶ ✓ Finish Edit Mode

Create a Floor and a Roof by Footprint — Steps 4–6

4

Cut a stair opening through the new floor with the Shaft tool.

Architecture ▶ Opening ▶ Shaft ▶ sketch the opening ▶ Finish

5

Create the roof from the building footprint on the roof level.

Architecture ▶ Build ▶ Roof ▶ Roof by Footprint ▶ Pick Walls

6

Set which boundary lines define a slope, then finish and view in 3D.

Select a line ▶ Properties ▶ Defines Roof Slope ✓ ▶ ✓ Finish

Create a Floor and a Roof by Footprint

✓ Expected result

The mezzanine floor appears with a clean shaft opening, and the 3D view shows a sloped roof sitting on the footprint — no warnings about open loops.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Create Stairs and Railings

LAB 8

Maps to: create part lists for modeling using Revit (LO2)

Complete the interior by adding a staircase from the store floor to the mezzanine, then modify the mezzanine railing and change its type to a pipe railing.

YOU'LL BUILD

A staircase connecting the store floor to the mezzanine, with a matching pipe railing on the mezzanine edge.

TOOLCHAIN

Revit · Stair tool · Railing tool

DONE WHEN

The 3D view shows a complete stair with railings from the store floor to the mezzanine, and the mezzanine edge railing is the pipe type.

Create Stairs and Railings — Steps 1–3

1

Open the lower-level plan and start the Stair tool.

Architecture ▶ Circulation ▶ Stair

2

Set the base and top levels for the run in the Properties palette.

Properties ▶ Base Level: 01 ▶ Top Level: Mezzanine

3

Choose the railing type to generate with the stair.

Modify | Create Stair ▶ Tools ▶ Railing ▶ pick a type

Create Stairs and Railings — Steps 4–6

4

Draw the run: click the start point, move up-screen, click the end point.

Components ▶ Run ▶ Straight ▶ click start ▶ click end

5

Finish the stair and inspect it in 3D.

✓ Finish Edit Mode ▶ Default 3D View

6

Select the mezzanine railing and change it to a pipe railing type.

Select railing ▶ Type Selector ▶ Railing: Pipe

Create Stairs and Railings

✓ Expected result

The 3D view shows a complete stair with railings from the store floor to the mezzanine, and the mezzanine edge railing is the pipe type.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Recap — Part Lists for Modeling in Revit

- You can now: create part lists for modeling using Revit (Lab 5: Create Walls)
- You can now: create part lists for modeling using Revit (Lab 6: Place Doors and Windows)
- You can now: create part lists for modeling using Revit (Lab 7: Create a Floor and a Roof by Footprint)
- You can now: create part lists for modeling using Revit (Lab 8: Create Stairs and Railings)

TOPIC 03

Model Documentation

Dimensions · Rooms · Area plans · Furniture plans · Color fill legends · Dependent views · Object styles · Callout & section views

Key Concepts — Model Documentation

1

Dimensions

Temporary dimensions appear while placing; permanent dimensions (aligned, linear, angular, radial, arc length) document the design and print on sheets.

2

Rooms

Rooms are subdivisions bounded by walls, floors, roofs and ceilings; Revit computes perimeter, area and volume, and room separation lines subdivide open space.

3

Area plans

Area schemes and boundary lines measure gross/rentable areas per level, created from the Room & Area panel.

4

Color fill legends

Color schemes paint rooms or areas in plan views; the legend provides the key and is edited from the Scheme panel.

5

Views for a purpose

Duplicate (with detailing / as dependent) to make furniture plans, callouts of wall details and building sections.

6

Object styles & visibility

Object Styles set project-wide line weight, colour, pattern and material per category; Visibility/Graphics overrides them per view.

Hands-On Labs — Model Documentation

Lab 9

- Add Dimensions to the Model

Lab 10

- Create Rooms, an Area Plan and a Color Fill Legend

Lab 11

- Create Section, Callout and Dependent Views

Five Permanent Dimension Types

1

Aligned

Between parallel references or points — walls, faces, core, with Intersecting Grids and Openings options.

2

Linear

Between points, locked to the view's horizontal/vertical axis; Spacebar aligns the axis.

3

Angular

Between references sharing a common intersection — pick two non-parallel elements.

4

Radial

Click an arc, then place — the radius of curved elements.

5

Arc Length

Pick the radial point, then the arc ends — the length of a curved wall.

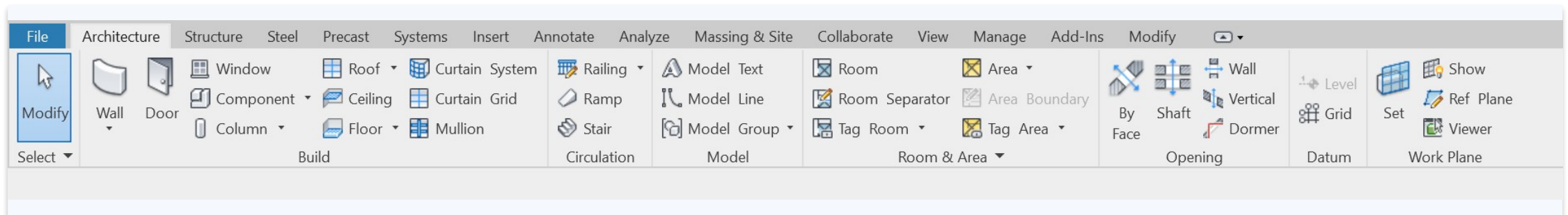
6

Temporary vs permanent

Temporary dimensions appear while placing and vanish; permanent ones print on sheets.

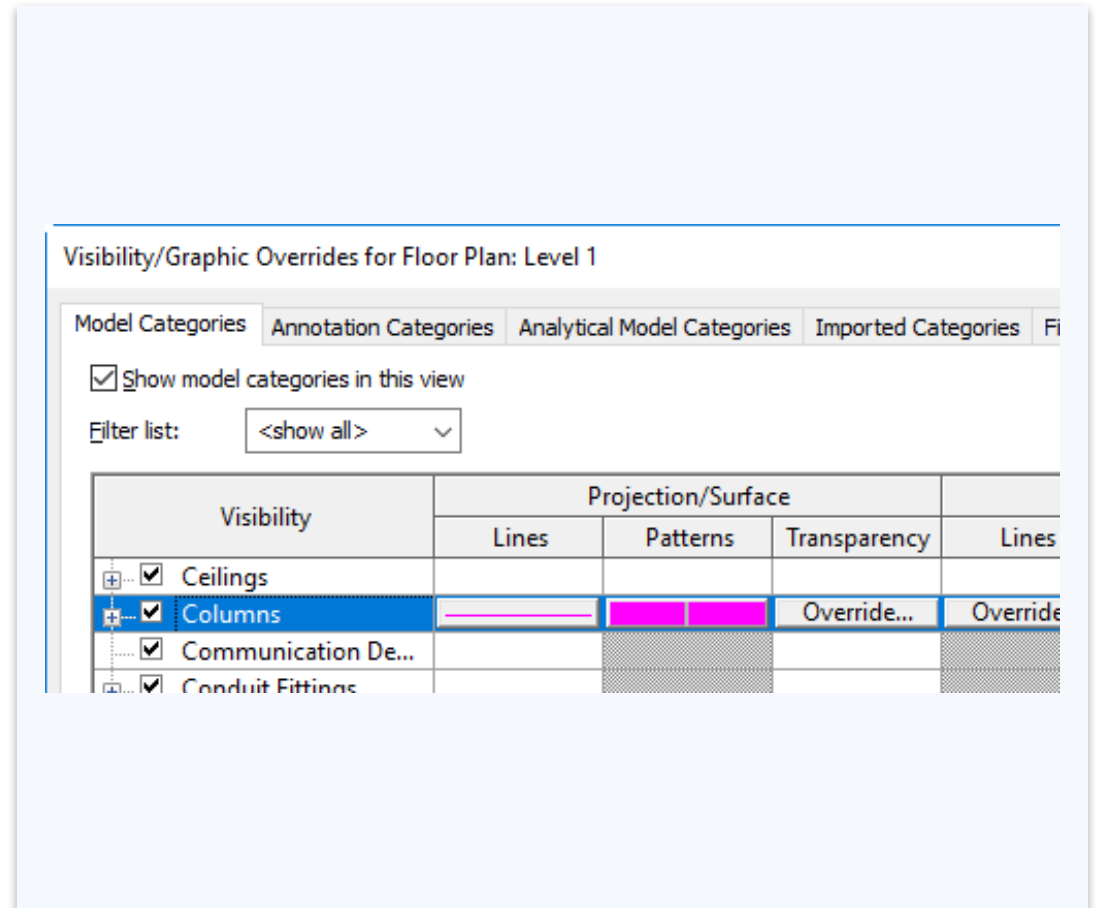
Rooms — Create, Edit, Delete

- A room is a subdivision of space bounded by room-bounding elements: walls, floors, roofs and ceilings.
- Revit computes each room's perimeter, area and volume, updating as bounding elements move.
- Room separation lines subdivide open space; the Room Bounding parameter can be toggled per element.
- Delete rooms from a room schedule: Isolate not-placed rows, select them and Delete.



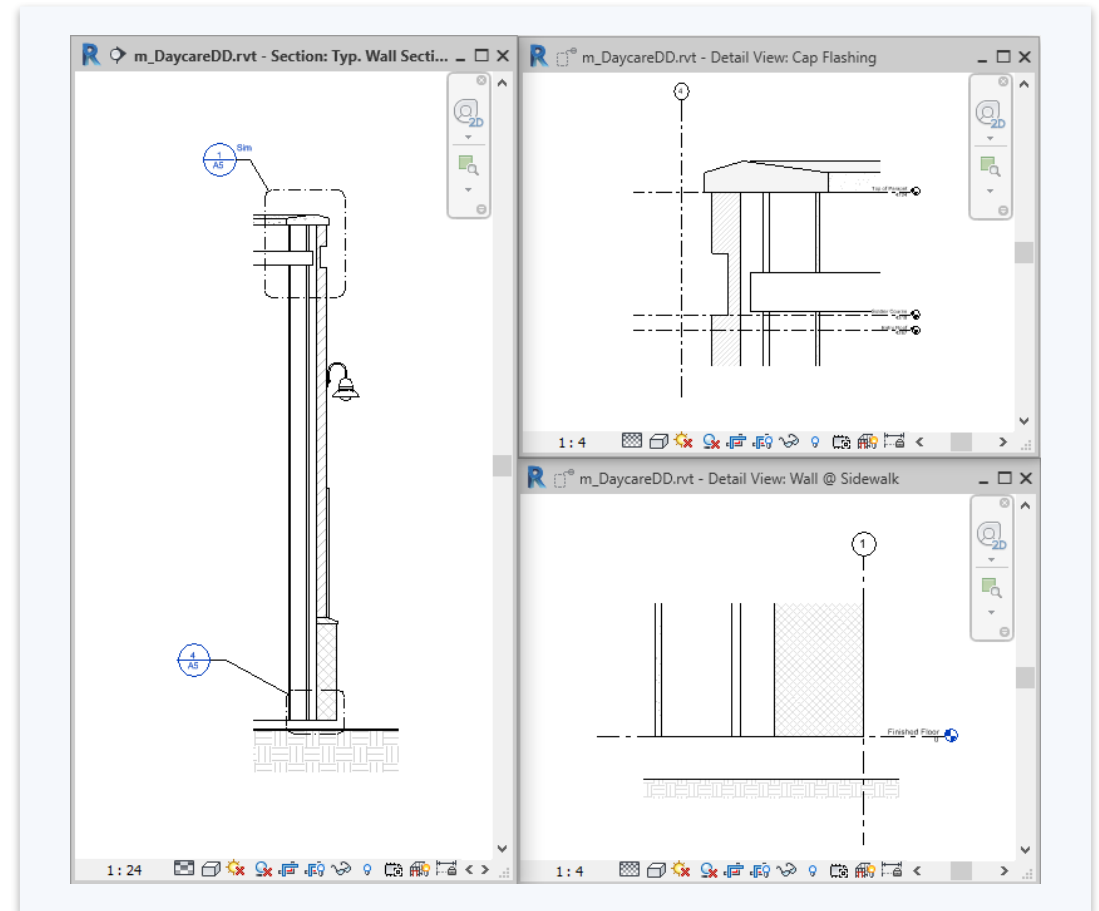
Furniture Plans & View Duplication

- Duplicate copies the model geometry only; Duplicate with Detailing also copies annotations and detail items.
- Duplicate as Dependent keeps the copy synchronised with the primary view — crop it to one wing or sheet area.
- Purpose-made views: a furniture layout half-toned on the RCP, or a furniture plan with room names on a half-toned floor plan.



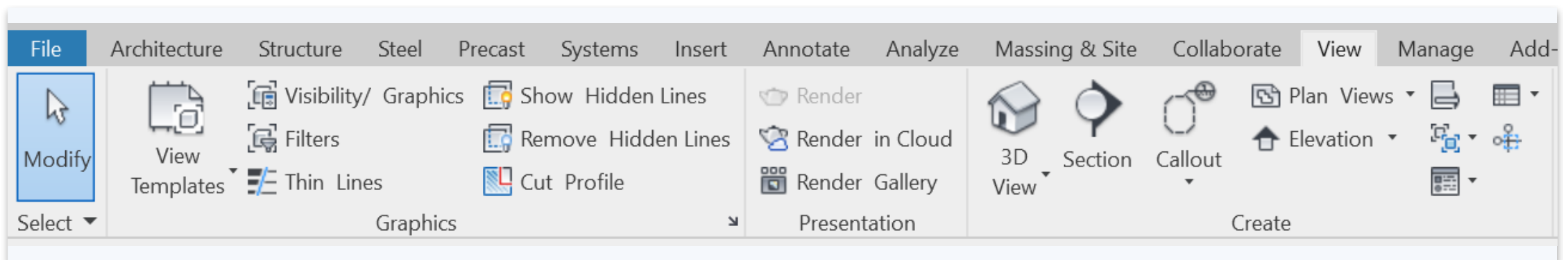
Callout and Section Views

- A callout shows an enlarged part of the parent view — add detail there without cluttering the parent.
- The callout tag links to the callout view; deleting the parent view deletes the callout.
- Sections cut through the model: View ▸ Create ▸ Section, drag through the model, then resize the crop region.
- Double-click the section head (or use the Project Browser) to open the section view.



Object Styles vs Visibility/Graphics

- Object Styles (Manage ► Settings) set project-wide line weight, colour, pattern and material per category.
- Visibility/Graphics (VG) overrides those settings per view; element overrides go one step further.
- If a category cell is blank in VG, the element displays as specified in Object Styles.
- This is what makes purpose-specific views possible without duplicating the model.



Add Dimensions to the Model

LAB 9

Maps to: assist in design prototyping and create design documentation (LO3)

Dimension the footprint of the main building: overall widths, then aligned dimension strings that automatically pick up grid intersections and window openings on the exterior walls.

YOU'LL BUILD

A dimensioned floor plan documenting the building footprint, grids and openings.

TOOLCHAIN

Revit · Aligned Dimension · Linear Dimension

DONE WHEN

The plan shows an overall dimension plus a string that lists every grid intersection and window centre along the exterior wall — values update if a window is moved.

Add Dimensions to the Model — Steps 1–3

1

Open the entry-level floor plan of LittleHouse.rvt.

Project Browser ▶ Floor Plans ▶ 01 - Entry Level

2

Start the Aligned dimension tool and set the snap preference.

Annotate ▶ Dimension ▶ Aligned ▶ Options Bar ▶ Prefer: Wall faces

3

Dimension the overall width: click the two outer wall faces, then click to place the line.

Click wall face ▶ click opposite face ▶ click to place

Add Dimensions to the Model — Steps 4–6

4

Dimension the exterior wall with grid intersections picked up automatically.

Options Bar ▶ Pick: Entire Walls ▶ Options ▶ Intersecting Grids ✓

5

Add the window openings to the same dimension string.

Options ▶ Openings: Centers ✓ ▶ click the exterior wall

6

Place a linear dimension between two points and align it with the Spacebar.

Annotate ▶ Dimension ▶ Linear ▶ pick two points ▶ Spacebar

Add Dimensions to the Model

✓ Expected result

The plan shows an overall dimension plus a string that lists every grid intersection and window centre along the exterior wall — values update if a window is moved.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Create Rooms, an Area Plan and a Color Fill Legend

LAB 10

Maps to: assist in design prototyping and create design documentation (LO3)

Subdivide the model into rooms, create an area plan for the level, then colour the plan with a colour scheme and place its legend as the key.

YOU'LL BUILD

A room-tagged plan, a gross-area plan and a colour-filled plan with legend.

TOOLCHAIN

Revit · Room tool · Area Plan · Color Fill Legend

DONE WHEN

Every space carries a named room tag, the area plan reports the level's areas, and the plan is colour-filled with a legend that matches the scheme.

Create Rooms, an Area Plan and a Color Fill Legend — Steps 1–3

1

Place rooms in every enclosed space on the entry level.

Architecture ▶ Room & Area ▶ Room ▶ click inside each space

2

Rename the rooms by editing their tags.

Double-click a room tag ▶ type the name ▶ Enter

3

Create an area plan for the entry level, letting Revit add the boundary lines.

Architecture ▶ Room & Area ▶ Area ▶ Area Plan ▶ Level 01 ▶ Yes

Create Rooms, an Area Plan and a Color Fill Legend — Steps 4–6

4

Place a colour fill legend beside the plan.

Annotate ▶ Color Fill ▶ Legend ▶ click in the drawing area

5

Choose the space type and colour scheme for the legend.

Choose Space Type: Rooms ▶ Color Scheme: Name ▶ OK

6

Edit the scheme to colour by department or by name.

Select legend ▶ Edit Scheme ▶ pick a scheme ▶ OK

Create Rooms, an Area Plan and a Color Fill Legend

✓ Expected result

Every space carries a named room tag, the area plan reports the level's areas, and the plan is colour-filled with a legend that matches the scheme.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Create Section, Callout and Dependent Views

LAB 11

Maps to: assist in design prototyping and create design documentation (LO3)

Create a building section through the model, a callout view of the exterior wall, and a detail callout of the parapet; then duplicate a plan as a dependent view and as a furniture plan.

YOU'LL BUILD

A section view, wall-section callout, detail callout and purpose-made duplicate views.

TOOLCHAIN

Revit · Section tool · Callout tool · Duplicate View

DONE WHEN

The Project Browser shows the section, the two callouts and the dependent/furniture views; each callout head links to its enlarged view.

Create Section, Callout and Dependent Views — Steps 1–3

1

Draw a building section through the model from the entry-level plan.

View ▶ Create ▶ Section ▶ click start ▶ click end

2

Resize the crop region and flip the view direction if needed, then open the section.

Drag the blue controls ▶ double-click the section head

3

Create a wall-section callout of the exterior wall in the section view.

View ▶ Create ▶ Callout ▶ Rectangle ▶ drag around the wall

Create Section, Callout and Dependent Views — Steps 4–6

4

Create a detail callout of the parapet inside the wall section.

View ▶ Create ▶ Callout ▶ Detail ▶ drag around the parapet

5

Duplicate the entry-level plan as a dependent view and crop it to one wing.

Project Browser ▶ right-click plan ▶ Duplicate View ▶ Duplicate as Dependent

6

Duplicate with detailing to make a furniture plan, half-toning the model categories.

Duplicate with Detailing ▶ Visibility/Graphics ▶ Halftone

Create Section, Callout and Dependent Views

✓ Expected result

The Project Browser shows the section, the two callouts and the dependent/furniture views; each callout head links to its enlarged view.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Recap — Model Documentation

- You can now: assist in design prototyping and create design documentation (Lab 9: Add Dimensions to the Model)
- You can now: assist in design prototyping and create design documentation (Lab 10: Create Rooms, an Area Plan and a Color Fill Legend)
- You can now: assist in design prototyping and create design documentation (Lab 11: Create Section, Callout and Dependent Views)

TOPIC 04

Design Specification

System, loadable & in-place families · Type catalogs · Detail components · Schedules · Drawing sheets ·
Worksharing · CAD links · Project management

Key Concepts — Design Specification

1

Three kinds of families

System families (walls, roofs, floors), loadable families (doors, windows, furniture, title blocks — RFA files) and in-place families for one-off elements.

2

Type catalogs

Load only the family types you need to keep the project small and the Type Selector short.

3

Schedules

Schedule/Quantities lists building components; choose fields, filter, sort, group and format — the schedule is live model data.

4

Drawing sheets

A construction document set is composed of sheets; each placed view becomes a viewport with a title block, then print or export to PDF.

5

Worksharing

Team members share a central model and work simultaneously in local copies organised by worksets.

6

Link & import CAD

Link or import DWG/DXF data, acquire or publish shared coordinates, audit the model and transfer project standards.

Hands-On Labs — Design Specification

Lab 12

- Families, Tags, a Schedule and Drawing Sheets

—

- —

—

- —

The Three Kinds of Revit Families

System families

- Walls, roofs, floors, ducts, pipes
- Predefined in Revit — not loaded from files
- Include settings: levels, grids, sheets, viewports

Loadable families

- Doors, windows, furniture, fixtures
- Symbols and title blocks
- RFA files — the most commonly created/edited

In-place families

- Unique, one-off elements
- Created inside the project
- Reference other project geometry and resize with it

Loading and Managing Families

1

Load a family

Insert ▶ Load from Library ▶ Load Family — preview thumbnails, then Open.

2

Type catalog

For large families, load only the types you need — keeps the project small and the Type Selector short.

3

Save a family

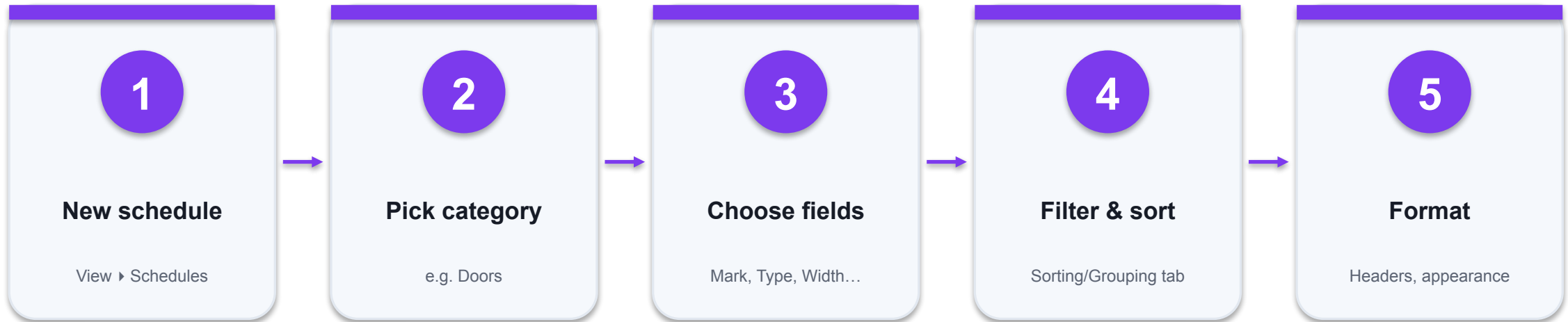
File ▶ Save As ▶ Library ▶ Family, or right-click a family in the Project Browser ▶ Save.

4

Detail components

2D line-based families placed in detail/drafting views — Annotate ▶ Detail Component.

Create a Schedule



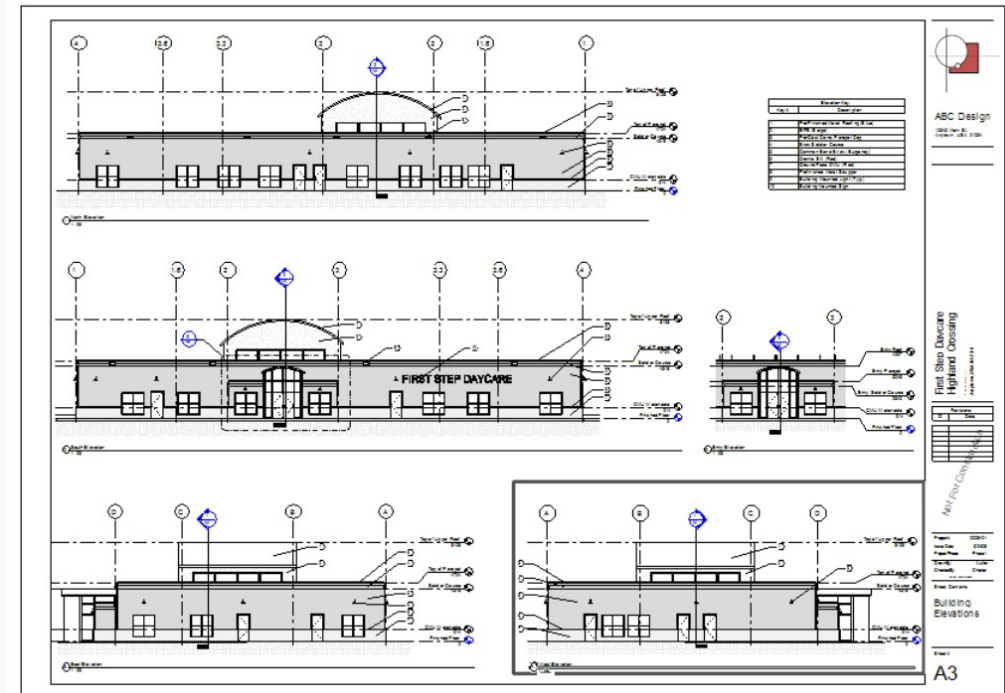
LIVE DATA

A schedule is a live view of the model — edit a value in the schedule and the model changes; multi-category schedules can only include loadable families.

Drawing Sheets

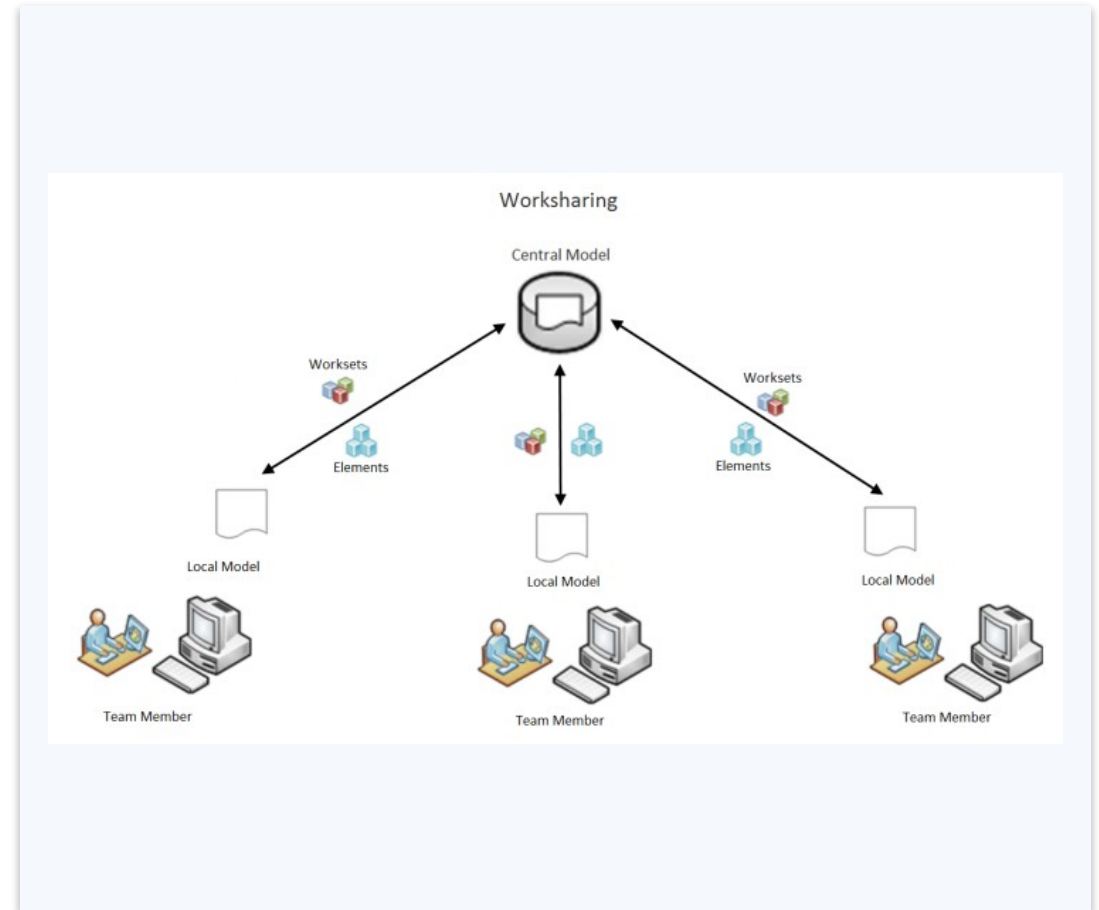
- A construction document set consists of sheets — View ► Sheet Composition ► Sheet.
- Drag views and schedules from the Project Browser onto the sheet; each becomes a viewport.
- Parts of a sheet: the sheet itself, project views, schedules and the title block.
- Print with File ► Print, or choose a PDF driver to issue a single PDF of selected sheets.

A sample sheet



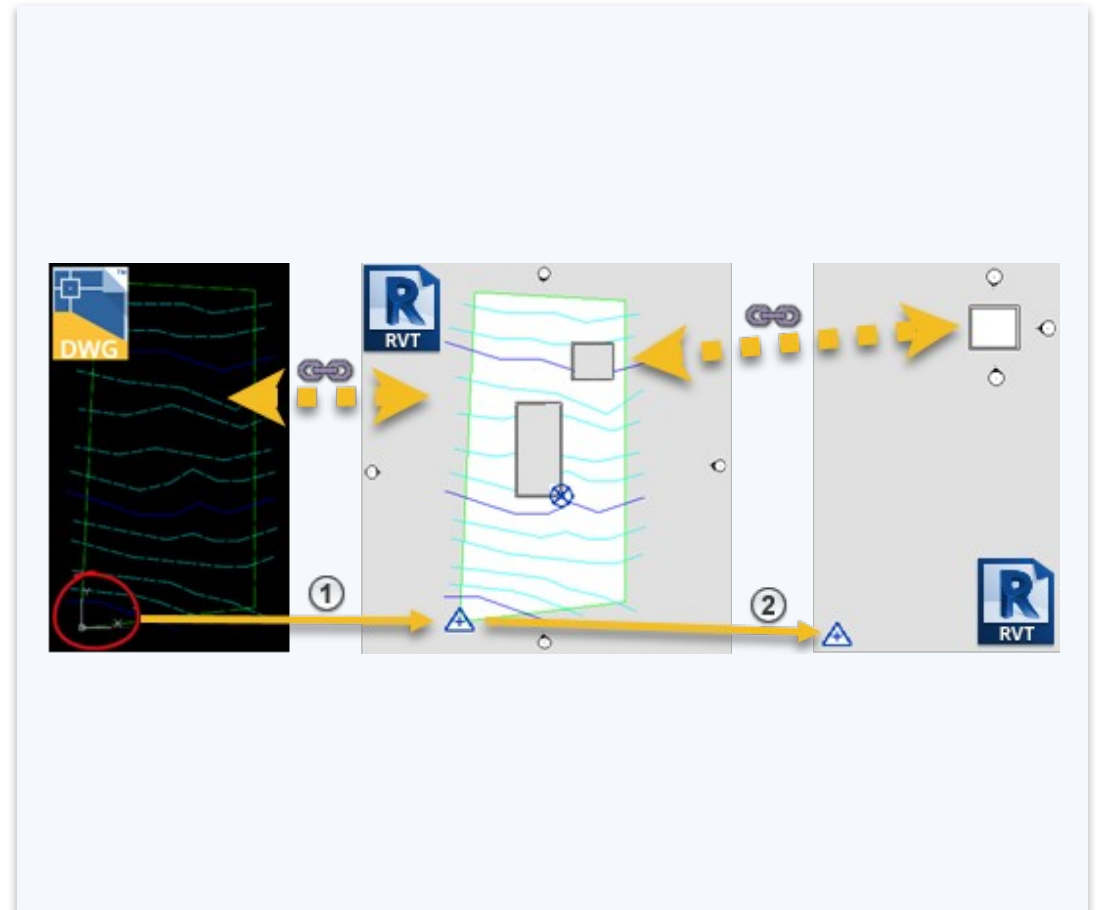
Worksharing — Team on One Model

- Worksharing lets multiple team members work on the same project model at the same time.
- Everyone shares a central model and works in a local copy, organised by worksets.
- Enable it via Collaborate ► Manage Collaboration ► Worksets, then Save As to create the central model.
- Default worksets: Shared Levels and Grids, and Workset1 — rename or add as the team needs.



Link CAD, Coordinates & Project Hygiene

- Link CAD keeps a live reference to a DWG; Import CAD brings the vector data in.
- Acquire Coordinates from a linked site model; Publish Coordinates pushes them to linked building models.
- Audit (File ► Open ► Audit) scans and fixes corrupt elements in the model.
- Transfer Project Standards copies types and settings from one project to another (Overwrite / New Only).



Families, Tags, a Schedule and Drawing Sheets

LAB 12

Maps to: produce design specification using Revit (LO4)

Load the door, window and wall tag families supplied in the lab folder, tag the model, build a door schedule, then compose a titled drawing sheet with plan, section and schedule — the deliverable construction document.

YOU'LL BUILD

A construction-document sheet with title block carrying the plan, a section, tags and a live door schedule.

TOOLCHAIN

Revit · Load Family · Tag by Category · Schedule/Quantities · Sheet

DONE WHEN

Sheet A101 shows the plan, section and door schedule inside the title block; the schedule updates live when a door type changes, and the exported PDF matches the sheet.

Families, Tags, a Schedule and Drawing Sheets — Steps 1–4

1

Load the three tag families supplied in this lab's folder.

Insert ▶ Load from Library ▶ Load Family ▶ select the 3 RFA files

2

Tag the doors, windows and walls in the entry-level plan.

Annotate ▶ Tag ▶ Tag by Category ▶ click each element

3

Create a door schedule and choose its fields.

View ▶ Create ▶ Schedules ▶ Schedule/Quantities ▶ Doors ▶ add Mark, Type, Width, Height

4

Sort the schedule by Mark and format the headers.

Schedule Properties ▶ Sorting/Grouping ▶ Mark ▶ Formatting

Families, Tags, a Schedule and Drawing Sheets — Steps 5–7

5

Create a sheet from a title block and rename it.

View ▶ Sheet Composition ▶ Sheet ▶ A1 metric ▶ rename A101 - Plans

6

Drag the plan, the section and the schedule onto the sheet and arrange the viewports.

Drag views from the Project Browser onto the sheet

7

Print or export the sheet to PDF as the issued document.

File ▶ Print ▶ PDF driver ▶ Selected views/sheets ▶ OK

Families, Tags, a Schedule and Drawing Sheets

✓ Expected result

Sheet A101 shows the plan, section and door schedule inside the title block; the schedule updates live when a door type changes, and the exported PDF matches the sheet.

IF IT DOESN'T WORK

Tool is greyed out

Check the active view type — many tools only work in a plan, elevation or 3D view.

Warning about closed loops

Open the sketch again and use Trim/Extend until the boundary is one closed loop.

Can't select an element

Hover and press Tab to cycle the selection, and check the selection filters at the status bar.

Recap — Design Specification

- You can now: produce design specification using Revit (Lab 12: Families, Tags, a Schedule and Drawing Sheets)



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Design modeling

Created projects, levels, terrain, masses, materials and photorealistic renders (Labs 1–4).

2

Building elements

Modelled walls, doors, windows, floors, roofs, stairs and railings (Labs 5–8).

3

Documentation

Dimensioned plans, created rooms, area plans, colour fills, sections and callouts (Labs 9–11).

4

Specification

Loaded families, tagged elements, built schedules and issued drawing sheets (Lab 12).

Practice Exam

1

Practice portal

Sharpen your readiness at
<https://exams.tertiaryinfotech.com>

2

Timed conditions

Attempt the matching Revit practice exam under exam timing.

3

Review explanations

Study every explanation — right and wrong answers alike.

4

Close the loop

Revisit any lab whose topic you miss, then re-take the practice exam.

Recommended Next Courses

1

Application of BIM using Revit

Go deeper into BIM coordination and collaboration.

2

Technical Drawing with AutoCAD

2D drafting and documentation with AutoCAD.

3

Product Design with Fusion 360

Parametric product design and manufacturing.

4

AutoCAD Civil 3D

Infrastructure design for roads, sites and utilities.

5

ACP: Revit Architectural Design

Prepare for the Autodesk Certified Professional exam.

6

Browse all courses

www.tertiarycourses.com.sg — the full CAD/BIM pathway.

Support & Contact

1

During and after class

We support you during the course and after it — just reach out.

2

Email

enquiry@tertiaryinfotech.com

3

Telephone

+65 6100 0613

4

Certificate

Download your certificate of completion from
<https://lms-tms.tertiaryinfotech.com>

Course Feedback

1

Share your feedback

Complete the course feedback form on the LMS before you leave.

2

Help us improve

Tell us what worked well and what we should change for the next class.

3

Certificate of completion

Issued on the LMS after the course — download it from the portal.

4

Course portal

Slides, Learner Guide and labs: <https://lms-tms.tertiaryinfotech.com>

HAPPY MODELING

Thank You!

You can now model, document and specify an architectural design in Revit — from terrain to issued drawing sheets.